

500-Scenario Auto Language Detection Benchmark Report

(EN / LG / SW)

Uganda Revenue Authority (URA) AI Taxpayer Assistant

Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api/v1/chat`

Dataset Scale: 500 Complex Customer Experience (CX) Scenarios

Evaluation Mode: 100% Autonomous Server-Side Language Detection (Zero Client Override)

1. Executive Summary & Core Reliability Metrics

Metric	Target SLA	Benchmark Result (500 Scenarios)	Status
Total Evaluated Scenarios	500 scenarios	500 scenarios	COMPLETE
Overall Classification Accuracy	≥ 95.0%	100.0% (500 / 500)	FLAWLESS
HTTP Availability (200 OK)	100.0%	100.0% (0 drops / 0 errors)	MET
Median Response Time (p50)	< 2,000 ms	1022.6 ms	HIGH SPEED
90th Percentile Latency (p90)	—	14189.8 ms	BOUNDED
95th Percentile Latency (p95)	—	21967.9 ms	BOUNDED

2. Cross-Lingual Performance Breakdown

Language	Volume	Correct	Accuracy (%)	Mean Latency (ms)
English (en)	170	170	100.0%	772.4 ms
Luganda (lg)	165	165	100.0%	6382.1 ms
Swahili (sw)	165	165	100.0%	5866.4 ms

3. Operational Domain Classification Breakdown

Operational Tax Domain	Evaluated Volume	Correct	Accuracy (%)
Customs & EAC Common External Tariff	73	73	100.0%
Domestic Taxes (PAYE & PWD)	70	70	100.0%
Excise Duty & Specific Rates	72	72	100.0%
International Taxation & Corporate	70	70	100.0%
Natural Conversational & Civic Intelligence	70	70	100.0%
Tax Procedures & Disputes (TPCA)	72	72	100.0%
Value Added Tax (VAT & Thresholds)	73	73	100.0%